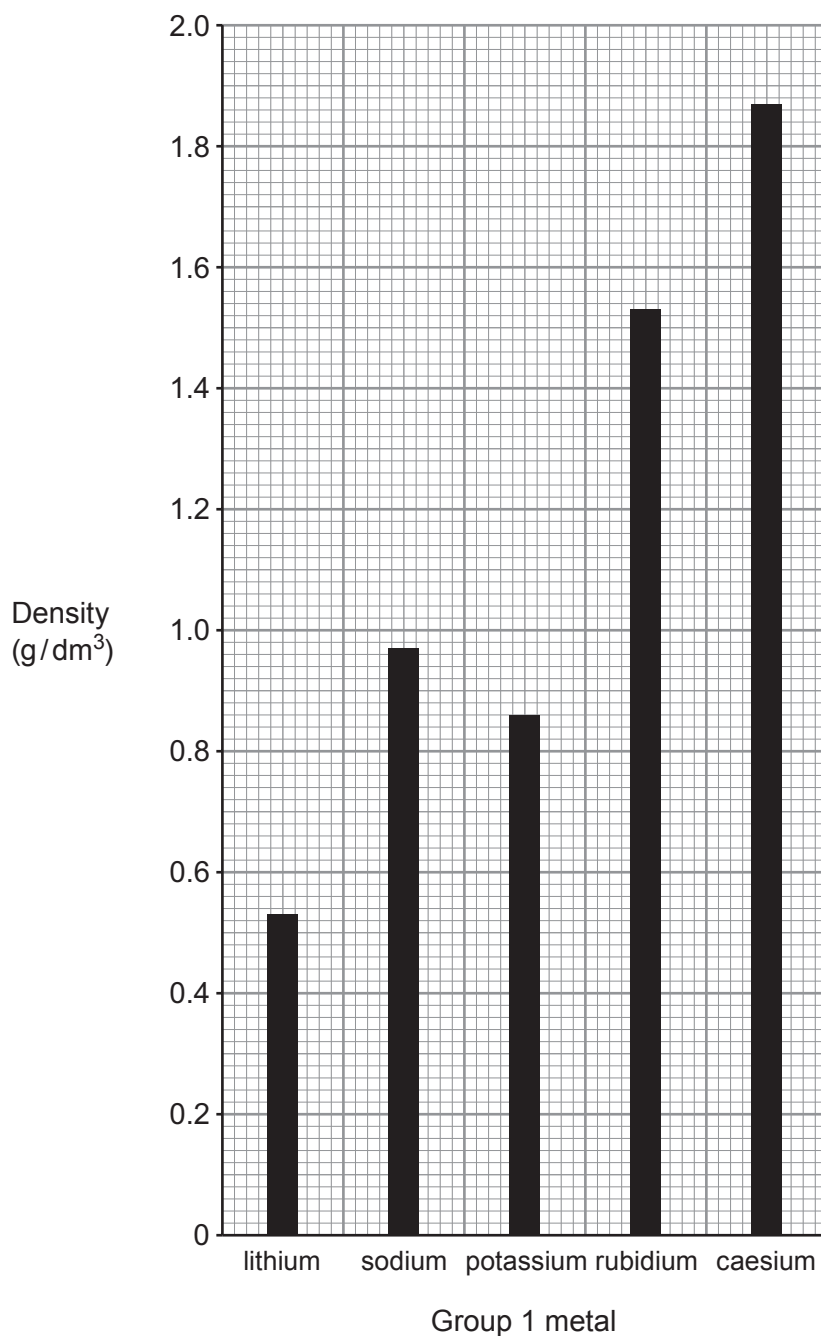


6. (a) The bar graph shows the densities of five Group 1 metals.



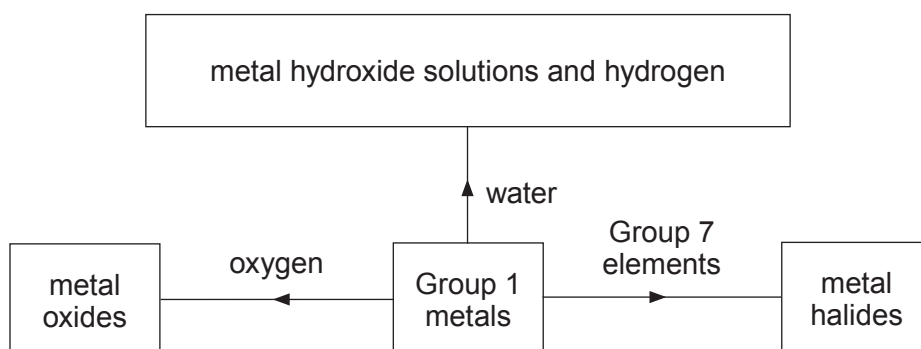
Draw a **straight** line on the bar graph showing the general trend in density of Group 1 metals. How does your trend line suggest that potassium and **not** sodium is the anomalous density value? [2]

.....

.....

.....

(b) The flow chart shows some reactions of Group 1 metals.



- (i) State and explain **one** similarity and **one** difference you would see when lithium and potassium are added separately to a trough of cold water. [4]

Similarity .....

.....

Difference .....

.....

- (ii) Write a balanced **symbol** equation to show the reaction between lithium and oxygen. [3]

.....

(iii)

| Group 1   | Group 7  |
|-----------|----------|
| lithium   | chlorine |
| sodium    | bromine  |
| potassium | iodine   |

The boxes show some of the elements in Group 1 and Group 7. Explain, in terms of electronic structure, why potassium and chlorine would react the most violently. [3]

.....

.....

.....

.....

.....

.....